Astana IT University

School of Software Engineering

PROBATIONARY PERIOD REPORT

Development of an Intelligent Assistant for Grading Diabetic Retinopathy Using Fundus Images

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Individual graduation project | Software Engineering

Analysis and design stage: 20 October 2026. English version: 24 October 2026.

Research prototype. Not intended for clinical diagnosis.

Astana, 2026

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1. Problem Statement and Project Proposal

Title: Development of an Intelligent Assistant for Grading Diabetic Retinopathy Using Fundus Images **Degree programme:** Software Engineering, Astana IT University **Student:** Roman Vladimirovich Kutbeyev, SE-2404 (individual project) **Supervisor:** Marzhan Kabdykalievna Turarova **Medical adviser:** to be confirmed **Version:** 0.2, 20 October 2026; English translation: 24 October 2026

1. Problem

Diabetic retinopathy (DR) is a microvascular complication of diabetes and a leading cause of vision loss among working-age adults. Early stages are asymptomatic, making regular fundus-image screening important for people with diabetes.

• Approximately 589 million adults aged 20-79 had diabetes in 2024; the projected number is 853 million by 2050. An estimated 43% are undiagnosed (IDF Diabetes Atlas, 11th edition, 2025).

• DR affects 22.27% of people with diabetes: approximately 103 million in 2020, projected to reach 160.5 million by 2045. Vision-threatening DR affects 6.17% (Teo et al., Ophthalmology, 2021).

Manual assessment requires qualified specialists whose numbers do not grow as quickly as the patient population. Automated screening systems, including IDx-DR/LumineticsCore and EyeArt, are proprietary and cannot be freely studied, reproduced or adapted. Open models trained on one dataset often lose performance on images from other cameras and clinics. The reliability of model confidence is also rarely reported.

2. Users

User	Task	Required output
Primary-care physician / endocrinologist	Assess referral to an ophthalmologist	Understandable recommendation and confidence
Ophthalmologist	Prioritise images and inspect model decisions	ICDR grade and heatmap
Screening operator	Upload an image and obtain a result	Simple interface and unsuitable-image message

These user groups and needs are assumptions to be checked with the medical adviser and practising clinicians (docs/meetings/).

3. Aim

Develop and evaluate a research prototype that grades DR from fundus images using the ICDR scale and supports referral assessment with calibrated confidence and a visual explanation. Evaluate reliability on independent external data.

4. Objectives

1. Review the domain, literature and existing automated DR screening systems. 2. Prepare APTOS 2019 and EyePACS for training/validation and IDRiD and Messidor-2 for external evaluation; perform EDA, preprocessing and leakage-free splitting. 3. Train a ResNet50/EfficientNet baseline; investigate ordinal regression with threshold selection, class imbalance, augmentation and preprocessing. 4. Evaluate internal/external performance and analyse deterioration across sources. 5. Calibrate probabilities using temperature scaling and evaluate ECE. 6. Generate Grad-CAM heatmaps and review them with the medical adviser. 7. Implement a web service: upload, grade, confidence, heatmap and referral-related output. 8. Test the system and prepare documentation and the thesis.

5. Expected Results

Deliverable	Criterion
Reproducible pipeline	README instructions produce comparable metrics
DR grading model	Internal-test QWK >= 0.85: research target, not a commitment
Evaluation report	Internal/external QWK, AUC, sensitivity at fixed specificity for referable DR, ECE
Explainability analysis	Grad-CAM review with the adviser
Web prototype	Upload-to-result workflow runs in Docker
Documentation	Requirements, architecture, ADRs, verification plan and thesis

6. Scope

Included: five ICDR classes; separate binary research label DR >= 2 without independent macular-oedema detection; public datasets; external evaluation; calibration; Grad-CAM; REST API and simple web interface. Image-quality assessment is optional if time permits.

Excluded: lesion segmentation; other eye diseases such as glaucoma and AMD; mobile application; domain adaptation; clinical validation; medical-device registration; health-information-system integration.

7. Constraints and Assumptions

• The prototype is for research and is not a medical device. Its output supports analysis and is not a diagnosis.

• Public datasets will be used under their licences; identifiable patient data will not be processed.

• Local development uses Apple Silicon and PyTorch MPS; larger experiments may require a rented GPU.

• Public labels may contain noise, which must be considered when interpreting metrics.

8. Stages

Period	Stage	Checkpoint
13-24 October 2026	Analysis, requirements, architecture, verification planning	School package
November-December 2026	Data, EDA, baseline	pre-defense-1
January-February 2027	Improvements, cross-dataset evaluation, calibration, Grad-CAM	pre-defense-2
March-April 2027	Service, testing, adviser-assisted error analysis	To be scheduled
April-May 2027	Thesis and final preparation	final-pre-defense

References

1. International Diabetes Federation. IDF Diabetes Atlas, 11th ed. Brussels, 2025. https://diabetesatlas.org/media/uploads/sites/3/2025/04/IDF_Atlas_11th_Edition_2025_Global-Factsheet.pdf 2. Teo Z.L. et al. Global prevalence of diabetic retinopathy and projection of burden through 2045: systematic review and meta-analysis. Ophthalmology, 2021, 128(11):1580-1591. 3. Wilkinson C.P. et al. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. Ophthalmology, 2003, 110(9):1677-1682.

The full reference list appears in the literature review.

Status Clarification: 20 October 2026

The proposal describes the intended thesis outcome. The current package contains analytical documents and mockups; no model or service has been implemented and no metrics have been measured. DR >= 2 defines the ground-truth binary label. The predicted flag uses p2+p3+p4 and a fixed threshold; it is not equivalent to argmax grade >= 2 and does not assess DME. Rules appear in requirements.md and verification-plan.md. Dataset usage conditions must be confirmed before downloading; public accessibility does not imply unrestricted use.

2. Evidence of Relevance and User Needs

Version 0.2, 20 October 2026. DIP-7. Desk research is complete; user validation remains incomplete.

Established Evidence

The Teo et al. meta-analysis estimated DR prevalence at 22.27% among people with diabetes. This is a global estimate, not a Kazakhstan-specific estimate [L01]. A deployment study in Thailand evaluated screening workflows, including referrals caused by ungradable images [L08]. ACCESS increased screening completion in a particular paediatric clinic; applicability to Kazakhstan has not been established [L10]. Sources and limitations appear in the literature review.

These findings support the importance of the problem and examining the full workflow, including image quality and subsequent actions. They do not establish local demand for this prototype.

Hypotheses to Validate

ID	Hypothesis	Method	Implication
H1	Clinicians benefit from grades 0-4 alongside a DR >= 2 flag	Discuss the mockup and three typical cases	Refine FR-03/04
H2	Grad-CAM helps inspect results	Compare discussion with/without a map	Refine FR-06; maps are not proof
H3	Operators need explicit invalid-file rejection	Walk through an invalid upload	Refine FR-01/02
H4	The workflow fits a local clinic	Discuss cameras, workload, responsibility	Revisit users and scope
H5	Users understand numerical confidence	Ask them to explain the value	Replace unclear labels

Plan for 3-5 Interviews

Plan an initial medical-adviser interview and 2-4 interviews with prospective users: an ophthalmologist, primary-care physician and screening operator. Identify repeat interviews explicitly; confirm with the supervisor whether they count. Dates, participants and a real-client agreement are not yet confirmed.

Questions cover current assessment, poor-quality images, referral information, consequential errors, positive/negative results, heatmap usefulness and permissible educational data. Do not collect identifiable patient information.

Recording Method

For each completed interview, use docs/meetings/template.md to record the date, role, documentation consent, questions/answers, observations, confirmed needs and requirement changes. No interviews have been conducted yet.

3. Literature Review

DIP-3. Version 0.2, 20 October 2026. Review of the domain and methods; no original experimental results.

Selection Method and Scope

Sources were checked on 20 October 2026 using PubMed, publisher, PMLR and arXiv pages. Search topics: diabetic retinopathy grading, external validation, retinal foundation models, calibration, explainability and prospective screening. Fifteen scientific publications are included; eight appeared in 2022-2025, within the preceding five years. Seven earlier works establish the grading scale, methods and datasets. Sources were selected through targeted searches; this is not a systematic meta-analysis.

Analysis uses available abstracts, article pages and method descriptions. Reading every full text or reproducing experiments is not claimed. Links identify primary publications or bibliographic records. Published findings are not results of this project.

Analysis Table

ID / year	Study and approach	Project use	Transfer limitation
L01 / 2021	Teo et al.; prevalence meta-analysis	Relevance	Global prevalence is not local demand
L02 / 2003	Wilkinson et al.; ICDR consensus	Five ordered grades	Macular-oedema scale is separate
L03 / 2016	Gulshan et al.; DL for referable DR	Task baseline and external testing	Binary detection does not establish five-class performance
L04 / 2017	Selvaraju et al.; Grad-CAM	CNN region attribution	Neither lesion labels nor causal proof
L05 / 2017	Guo et al.; temperature scaling	Held-out calibration	No guarantee on another camera
L06 / 2019	Tan and Le; EfficientNet	Compact CNN baseline	ImageNet efficiency does not guarantee DR performance
L07 / 2018	Porwal et al.; IDRiD	Independent evaluation source	Distinguish subtasks and label availability
L08 / 2022	Ruamviboonsuk et al.; prospective Thai screening	Workflow and rejection analysis	Different setting, equipment and protocol
L09 / 2022	Automated multidimensional platform; multicentre retrospective evaluation	Gradability and image limits	Retrospective testing does not replace deployment
L10 / 2024	Wolf et al.; ACCESS randomised trial	Screening completion beyond AUC	One paediatric centre; no proof of local demand
L11 / 2024	Autonomous AI in underserved populations; deployment study	Access and group differences	Observational association is not causation
L12 / 2024	Dai et al.; DeepDR Plus progression prediction	External cohorts and alternative endpoint	Time-to-progression is outside scope
L13 / 2023	Zhou et al.; RETFound self-supervision	CNN alternative and label efficiency	Resources and pretraining overlap complicate testing
L14 / 2025	Data-efficient strategy; RETFound-DE	Data-efficient alternatives	Synthetic-data generation is outside scope
L15 / 2025	Engelmann and Bernabeu; RETFound-Green	Foundation-model cost/compactness	Reported gains belong to the authors' protocol

Synthesis

ICDR [L02] defines ordered output, but mtmDR detection differs from five-grade classification. Report QWK and per-class errors separately from binary AUROC and sensitivity/specificity. Unlike clinical screening systems, this prototype does not assess macular oedema or provide a complete referral decision.

Deployment studies [L08-L11] show the importance of workflows, gradability and follow-up. They motivate studying the scenario but do not replace 3-5 interviews with actual prospective users.

RETFound and economical extensions [L13-L15] broaden alternatives beyond CNNs. Given time limits, start with a reproducible compact baseline [L06]. Foundation models remain a comparison direction; changing models requires measured improvement under one protocol without pretraining/test leakage.

A confident model can be wrong. Evaluate calibration [L05] separately from classification and again externally. Use Grad-CAM [L04] for error analysis and adviser discussion without assuming clinical validity.

Research Questions

RQ1: How do QWK and binary DR >= 2 detection change from internal to independent external data? RQ2: Does temperature scaling improve NLL/ECE, and does the improvement persist externally? RQ3: Which errors and artifacts emerge through Grad-CAM analysis? These questions define a testable educational contribution without claiming a new clinical system.

Bibliography

[L01] Teo Z.L. et al. Global Prevalence of Diabetic Retinopathy and Projection of Burden through 2045: Systematic Review and Meta-analysis. Ophthalmology, 2021, 128(11):1580-1591. https://pubmed.ncbi.nlm.nih.gov/33940045/

[L02] Wilkinson C.P. et al. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. Ophthalmology, 2003, 110(9):1677-1682. DOI 10.1016/S0161-6420(03)00475-5. https://pubmed.ncbi.nlm.nih.gov/13129861/

[L03] Gulshan V. et al. Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. JAMA, 2016, 316(22):2402-2410. DOI 10.1001/jama.2016.17216. https://pubmed.ncbi.nlm.nih.gov/27898976/

[L04] Selvaraju R.R. et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV, 2017; preprint 2016. https://arxiv.org/abs/1610.02391

[L05] Guo C. et al. On Calibration of Modern Neural Networks. ICML, PMLR 70:1321-1330, 2017. https://proceedings.mlr.press/v70/guo17a.html

[L06] Tan M., Le Q. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ICML, PMLR 97:6105-6114, 2019. https://proceedings.mlr.press/v97/tan19a.html

[L07] Porwal P. et al. Indian Diabetic Retinopathy Image Dataset (IDRiD): A Database for Diabetic Retinopathy Screening Research. Data, 2018, 3(3):25. DOI 10.3390/data3030025. https://www.mdpi.com/2306-5729/3/3/25

[L08] Ruamviboonsuk P. et al. Real-time diabetic retinopathy screening by deep learning in a multisite national screening programme: a prospective interventional cohort study. The Lancet Digital Health, 2022. https://pubmed.ncbi.nlm.nih.gov/35272972/

[L09] Zhang G. et al. Automated multidimensional deep learning platform for referable diabetic retinopathy detection: a multicentre, retrospective study. BMJ Open, 2022, 12:e060155. DOI 10.1136/bmjopen-2021-060155. https://pubmed.ncbi.nlm.nih.gov/35902186/

[L10] Wolf R.M. et al. Autonomous artificial intelligence increases screening and follow-up for diabetic retinopathy in youth: the ACCESS randomized control trial. Nature Communications, 2024, 15:421. DOI 10.1038/s41467-023-44676-z. https://www.nature.com/articles/s41467-023-44676-z

[L11] Autonomous artificial intelligence for diabetic eye disease increases access and health equity in underserved populations. npj Digital Medicine, 2024, 7:196. DOI 10.1038/s41746-024-01197-3. https://www.nature.com/articles/s41746-024-01197-3

[L12] Dai L. et al. A deep learning system for predicting time to progression of diabetic retinopathy. Nature Medicine, 2024, 30:584-594. DOI 10.1038/s41591-023-02702-z. https://pubmed.ncbi.nlm.nih.gov/38177850/

[L13] Zhou Y. et al. A foundation model for generalizable disease detection from retinal images. Nature, 2023, 622:156-163. DOI 10.1038/s41586-023-06555-x. https://www.nature.com/articles/s41586-023-06555-x

[L14] A data-efficient strategy for building high-performing medical foundation models. Nature Biomedical Engineering, 2025. DOI 10.1038/s41551-025-01365-0. https://www.nature.com/articles/s41551-025-01365-0

[L15] Engelmann J., Bernabeu M.O. Training a high-performance retinal foundation model with half-the-data and 400 times less compute. Nature Communications, 2025, 16:6862. DOI 10.1038/s41467-025-62123-z. https://www.nature.com/articles/s41467-025-62123-z

Publication Data, Metrics and Findings

These are published findings, not project results. Numerical values absent from the checked descriptions are not reconstructed from memory.

ID	Data / design	Evaluation	Finding and limitation
L01	Meta-analysis of people with diabetes	Prevalence and 95% CI	DR 22.27%; no local-demand estimate
L02	International expert consensus	Classification agreement	Five-grade scale, not an ML benchmark
L03	Labelled fundus photos; EyePACS-1/Messidor-2 evaluation	Sensitivity, specificity, AUC	DL detects referable DR; binary task
L04	Image recognition tasks	Localisation and qualitative maps	Gradient maps indicate CNN attention; no clinical endpoint
L05	Image/document classification	Calibration including ECE	Temperature scaling improves calibration on studied tasks
L06	ImageNet and transfer tasks	Accuracy, parameters, compute	Joint depth/width/resolution scaling works under authors' protocol
L07	IDRiD images with grading/subtask labels	Dataset, annotation and task description	Research dataset, not our model result
L08	7,651 eligible Thai screening participants	Accuracy, sensitivity, specificity for vision-threatening DR	Prospective workflow evaluation; endpoint broader than DR >= 2
L09	Multicentre retrospective data; five classifiers	Accuracy, F1, sensitivity, specificity, AUROC, AUPRC	Quality and screening components evaluated; no direct prototype comparison
L10	ACCESS: 81 intervention and 83 control participants	Screening completion within six months	100% versus 22% in that study, not model accuracy
L11	Integrated health-system deployment; >17,000 adults with diabetes	Screening adherence and group differences	AI use associated with access gains; confounding possible
L12	Pretraining and longitudinal progression cohorts	C-index and integrated Brier score	Time to progression, not current grade
L13	Self-supervision on about 1.6 million CFP/OCT images and downstream tasks	Downstream performance, label efficiency, external testing	Useful representations; no guaranteed gain on our split
L14	Synthetic/real fundus data; nine public datasets	Downstream performance and label needs	Data-efficient pretraining reduces real-data needs in the study
L15	75,000 public pretraining images and external downstream sets	Performance and training/inference resources	Compact RETFound-Green reduces resources under authors' protocol; no local measurement

4. Existing System Comparison

DIP-4. 20 October 2026. Three systems are compared against eight criteria using specific FDA documents. Characteristics apply to the cited regulatory versions.

Criterion	IDx-DR [S1]	EyeArt [S2]	AEYE-DS [S3]
Comparison basis	De Novo DEN180001, 2018	510(k) K200667, 2020	510(k) K240058, 2024
Intended use	Automated more-than-mild DR detection	mtmDR and vision-threatening DR detection	Automated mtmDR detection
Target population	Adults with diabetes without previously diagnosed DR	Adults with diabetes, subject to IFU restrictions	Adults with diabetes without previously diagnosed DR
Output level	mtmDR screening result	Screening endpoints	mtmDR screening result
Five ICDR grades as user output	Not stated in reviewed intended use	Not stated in reviewed intended use	Not stated in reviewed intended use
Equipment	Topcon NW400	Cameras specified in document	Topcon NW400 and Optomed Aurora
Image gradability	Image-quality checks	Image-quality checks	Image-quality checks
Pre-market evaluation	Clinical evidence in De Novo	Clinical/equivalence evidence in 510(k)	Clinical/equivalence evidence in 510(k)

For five-grade output, absence of a statement is not proof of absence from all versions. Open weights, pricing, Kazakhstan availability and user-facing calibrated probabilities were not established from these documents.

Design Implications

Each system has a defined intended use, population and compatible equipment. A student prototype accepting arbitrary uploads cannot inherit its clinical conclusions. File-decoding checks are not clinical gradability assessment.

The project targets research reproducibility: five classes, separate DR >= 2 label, external evaluation, calibration and Grad-CAM. These are planned properties, not demonstrated advantages. Unlike regulatory mtmDR definitions, the binary label does not independently detect DME. Trial sensitivity/specificity values are not ranked because populations, reference standards and endpoints differ.

Primary Sources

[S1] FDA. De Novo classification request for IDx-DR, DEN180001: intended use, restrictions, technical/clinical evaluation.
https://www.accessdata.fda.gov/cdrh_docs/reviews/DEN180001.pdf

[S2] FDA. EyeArt 510(k) summary, K200667: indications, system and clinical study.
https://www.accessdata.fda.gov/cdrh_docs/pdf20/K200667.pdf

[S3] FDA. AEYE-DS 510(k) summary, K240058: intended use and cameras.
https://www.accessdata.fda.gov/cdrh_docs/pdf24/K240058.pdf

5. Technology Comparison

DIP-5. Version 0.2, 20 October 2026. Four decisions with three alternatives each. Suitability assessments are engineering judgements, not benchmarks. Dependency versions will be pinned after a trial run; training code is not yet available.

Decision	Alternative	Strength	Constraint	Selection
ML framework	PyTorch	Eager execution, Apple MPS, Grad-CAM gradients [T1]	MPS operation support needs testing	Selected
ML framework	TensorFlow/Keras	High-level training, SavedModel [T2]	Different ecosystem; Mac testing	Reserve
ML framework	JAX	Autodiff and JIT [T3]	More custom training/serving code	Not selected
Backbone	EfficientNet-B0	Compact CNN and scaling [L06]	Downscaling may remove small features	First baseline
Backbone	ResNet50	Standard CNN and pretrained weights [T4]	Different parameter/speed trade-off	Controlled comparison
Backbone	RETFound	Retinal pretraining [L13]	Size and data-overlap checks	Later alternative
Interface/service	FastAPI + HTML/JS	Explicit API contract, OpenAPI [T5]	Separate UI implementation	Selected
Interface/service	Streamlit	Rapid research interface [T6]	Execution/session model	Demo reserve
Interface/service	Gradio	ML demo components [T7]	Less control under chosen API/UX approach	Reserve
Inference	PyTorch eager	Same weights/preprocessing; gradients	ML environment in service	First version
Inference	ONNX Runtime	Multiple device providers [T8]	Export/equivalence; separate Grad-CAM	After baseline
Inference	TorchScript	PyTorch execution artifact	Officially deprecated [T9]	Not selected

Decision Rationale

Priorities are single-author reproducibility, the available Mac, Grad-CAM and limited time. Start with PyTorch, EfficientNet-B0 and FastAPI without another inference runtime. Compare ResNet50 on the same data/resources. Adopt ONNX only after confirming maximum probability difference <= 1e-4 on a control set and a measurable time/memory improvement. ONNX requires a separate PyTorch Grad-CAM path with matching weights.

Documentation Sources

[T1] PyTorch, MPS: https://docs.pytorch.org/docs/stable/notes/mps.html

[T2] TensorFlow, Keras: https://www.tensorflow.org/guide/keras

[T3] JAX: https://docs.jax.dev/en/latest/

[T4] Torchvision, ResNet50: https://docs.pytorch.org/vision/stable/models/generated/torchvision.models.resnet50.html

[T5] FastAPI: https://fastapi.tiangolo.com/features/

[T6] Streamlit execution model: https://docs.streamlit.io/develop/concepts/architecture

[T7] Gradio: https://www.gradio.app/guides/quickstart

[T8] ONNX Runtime: https://onnxruntime.ai/docs/execution-providers/

[T9] TorchScript: https://docs.pytorch.org/docs/stable/jit.html

Scientific references L06 and L13 appear in the literature review.

6. Requirements Specification

Version 0.2, 20 October 2026. DIP-8. Draft requirements for a research prototype; user validation is pending. Owner: Roman Vladimirovich Kutbeyev.

Workflow and Terminology

An operator uploads one colour fundus image from an authorised de-identified dataset. The system validates and preprocesses the file and returns a research assessment. ICDR grades: 0, no apparent DR; 1, mild; 2, moderate; 3, severe non-proliferative; 4, proliferative [L02]. DR >= 2 defines this project's binary research label. It excludes independent diabetic macular oedema detection and is not the complete clinical referral decision.

Functional Requirements

ID	Priority	Requirement	Acceptance criterion
FR-01	Must	Upload one JPG/PNG, up to 20 MiB	Valid RGB/grayscale images decode to RGB; 20 MiB + 1 byte is rejected; a valid extension alone is insufficient
FR-02	Must	Validate decoding and dimensions	Empty/corrupt file: 422; unsupported format: 415; over 40 million pixels or 20 MiB: 413; no model invocation
FR-03	Must	Grade 0-4	Integer grade in 0..4; five finite probabilities in [0,1], sum 1 within 1e-5; grade = argmax
FR-04	Must	Binary research flag	score = p2+p3+p4; flag = score >= threshold; response includes threshold and its version; do not replace rule with grade >= 2
FR-05	Must	Explain confidence	Display max(p), calibration status and 'model estimate, not probability of being healthy'; explicitly mark uncalibrated output
FR-06	Should	CNN Grad-CAM	Overlay on the same image after reversing geometric transforms; identify selected class; map failure preserves grade and produces a warning
FR-07	Must	Research-use statement	Warning on upload/result screens; a negative flag does not promise absence of other diseases
FR-08	Must	State management	Upload, processing, result and error states visible; prevent resubmission during processing; hide old output after new file selection
FR-09	Must	Result identification	API returns model_version, preprocessing_version, request_id and threshold; experiment report retains the same versions
FR-10	Could	Automatic gradability assessment	Implement only after independent quality labels and a threshold are defined; no clinical gradability claim beforehand

Non-functional Requirements

ID	Priority	Requirement	Acceptance criterion
NFR-01	Must	Reproducibility	Retain split manifest, seed, data SHA-256, dependency versions and weights; repeated evaluation of identical weights differs in QWK by <= 0.001
NFR-02	Must	Data minimisation	No images in database/logs; delete temporary files after response and exceptions; test temporary directory
NFR-03	Must	Resilience	Missing model: 503; internal error: 500 without path/stack disclosure; errors include request_id
NFR-04	Should	Response time	Target CPU p95 <= 10 s over 100 sequential images after 10 warm-ups, batch=1, 384 px; record CPU, RAM, OS, threads and version before measurement; no measurements yet
NFR-05	Must	Test-data isolation	No patient_id or duplicate overlap between train, tuning, calibration and test; disclose unknown patient identity
NFR-06	Must	Verifiable performance	Publish QWK, 5x5 confusion matrix, AUROC, sensitivity/specificity, ECE and confidence intervals under the plan; QWK 0.85 is a target
NFR-07	Should	Accessibility	Keyboard-accessible workflow, labelled file input, text errors; no horizontal scrolling at 375 px
NFR-08	Must	Execution restrictions	Never execute uploads or use filenames as paths; check byte and decoded-pixel limits before inference

Draft API Contract

POST /api/v1/predict, multipart field image. Success 200: request_id, grade, probabilities[5], referable_score, referable_threshold, referable_flag, confidence, calibration_status, model_version, preprocessing_version, optional heatmap, warnings[]. Errors 413/415/422/503/500 contain error.code, error.message and request_id. Mockup responses are not outputs of a working model.

Traceability

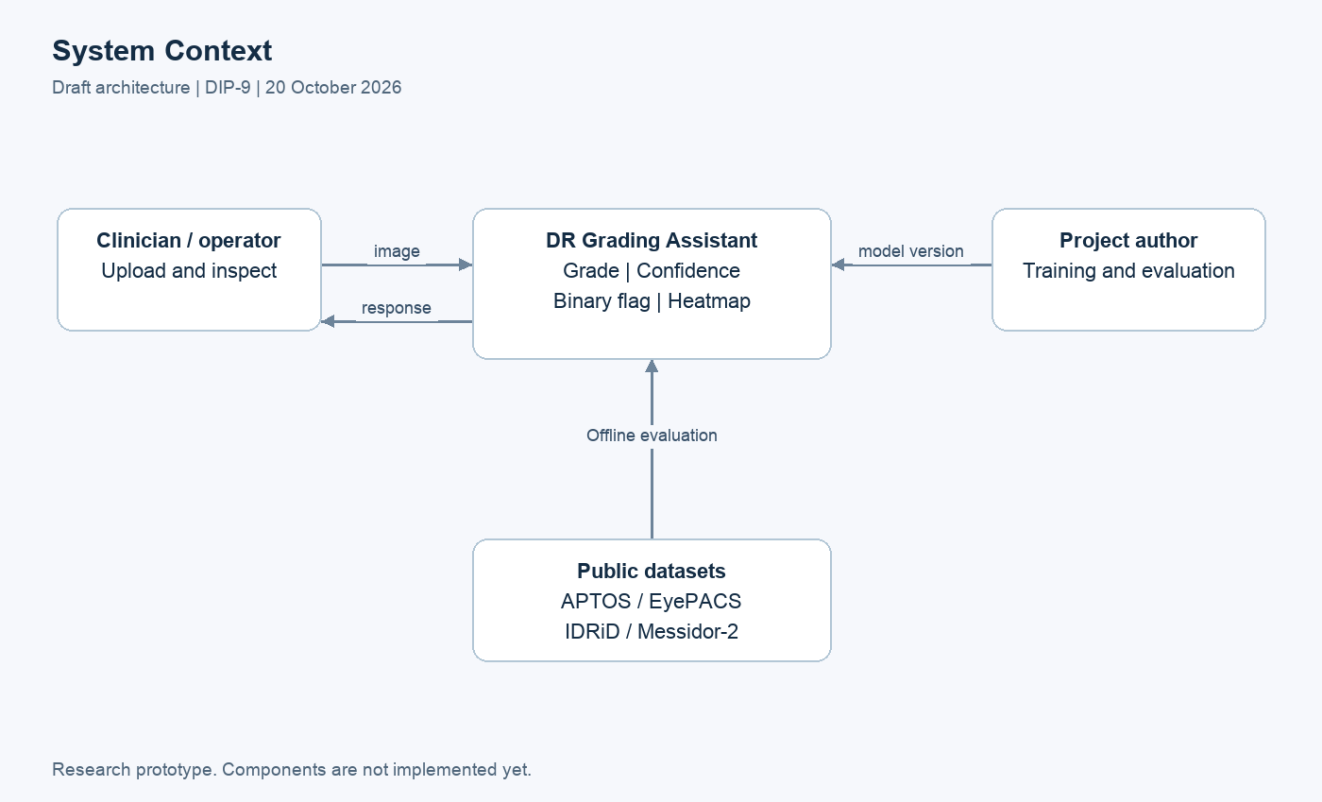
FR-01/02/08 -> UC-01 upload and UC-03 error -> T01-T03/T06. FR-03/04/05/09 -> UC-02 result -> T04/T05/T09. FR-06 -> T07. FR-07 -> T08. NFR-01/05/06 -> E01-E05. NFR-02/03/08 -> T02/T03/T06/T10. NFR-04 -> P01. NFR-07 -> U01. Test descriptions appear in the verification plan.

7. Architecture and Diagrams

DIP-9. Version 0.2, 20 October 2026. Draft architecture; software components have not been implemented.

System Boundary

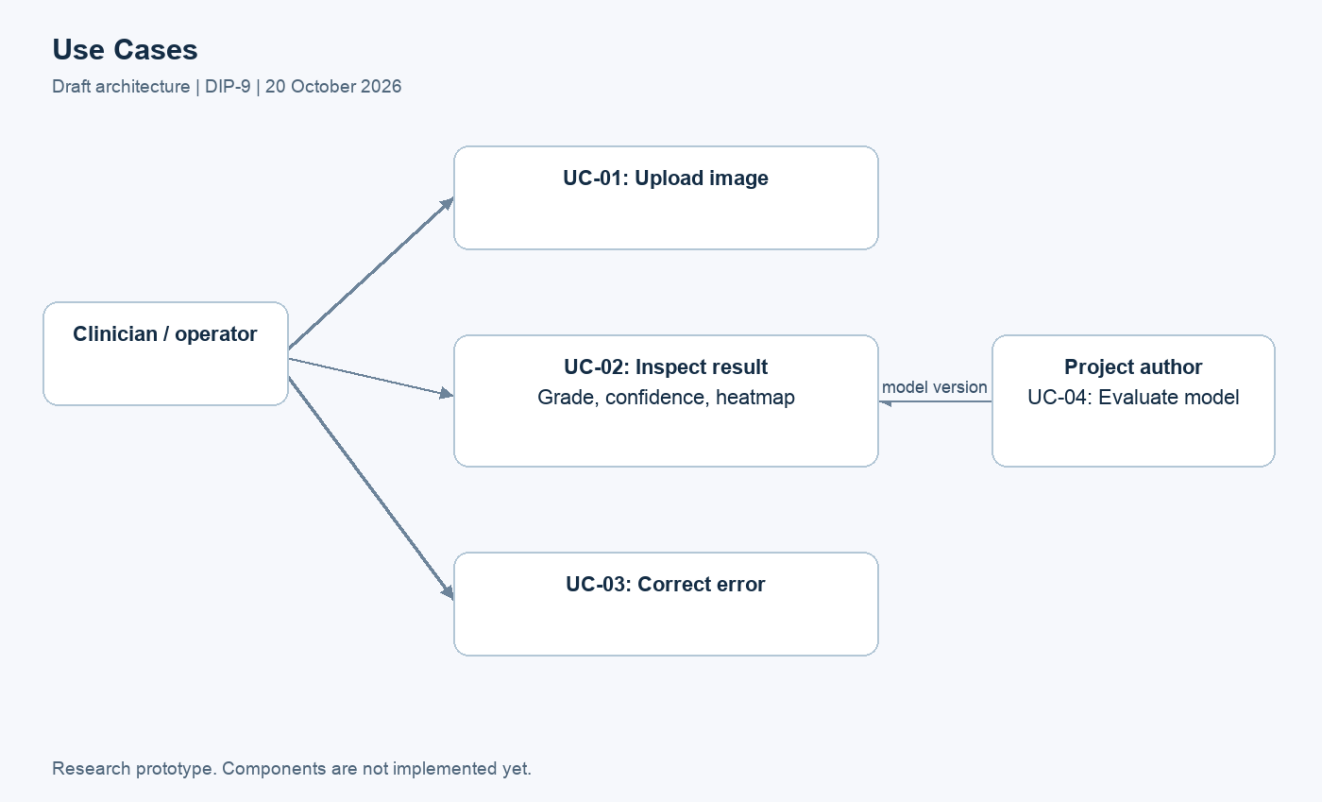
The system accepts one image and returns a research assessment. The author performs training/evaluation offline; user uploads do not trigger model updates. Health-information systems, patient registration and medical-record databases are excluded. The context does not imply access to clinical data.



System context

Use Cases

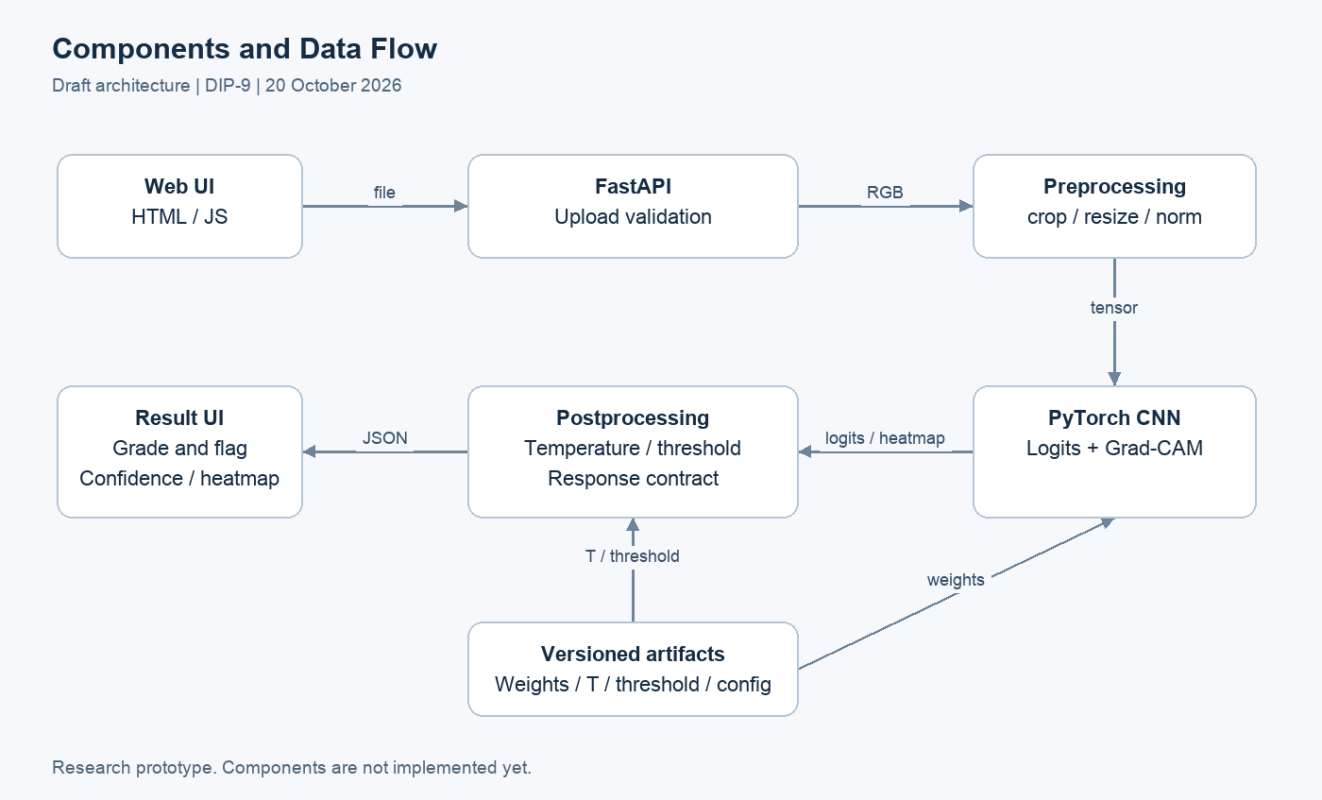
UC-01: select an allowed file, pass validation and wait for output. UC-02: inspect grade, binary flag and its limitation, confidence and heatmap. UC-03: correct a file after an explicit error without a stale result. UC-04: the author prepares data, evaluates and releases a model version. User interviews may change these scenarios.



Use cases

Components

Web UI sends a multipart request to FastAPI. Validation checks bytes, format and pixel limits. Preprocessing converts RGB to a tensor using the same configuration as training. The CNN returns logits; temperature scaling converts them into calibrated probabilities. Postprocessing computes grade=argmax and score=p2+p3+p4, comparing score with a fixed threshold. Grad-CAM requires activations/gradients from the same CNN. Map failure does not invalidate a correct main response.



Components

Weights, temperature, threshold, class order and preprocessing rules form one versioned bundle with checksums. Keep images only in memory or a temporary directory with cleanup. Logs retain request_id, version, duration and error code without image contents. Concurrent training and inference in one process are not planned.

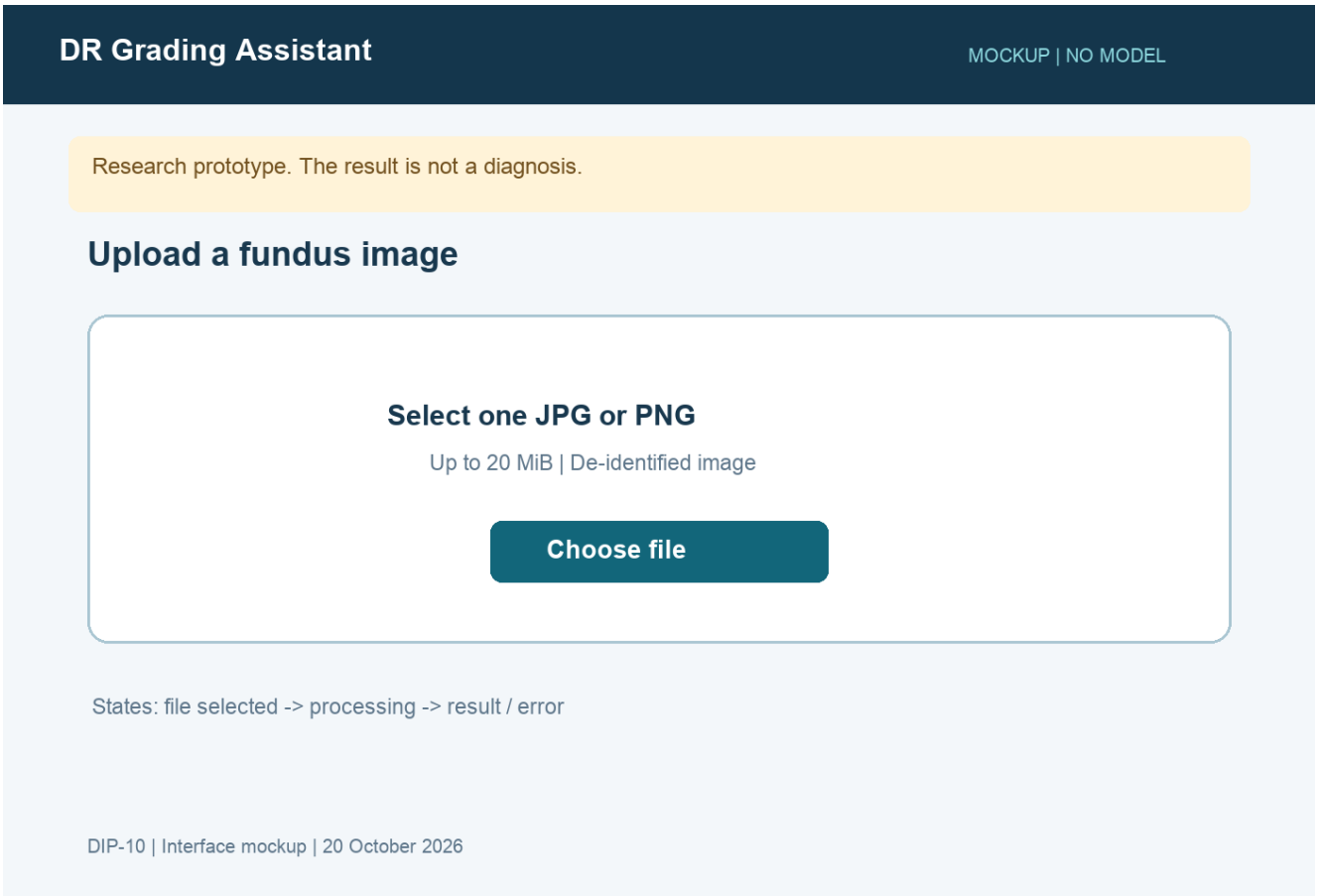
Editable Sources and Exports

Mermaid .mmd files and editable geometry JSON are in diagrams/src/. PNG/SVG exports are in diagrams/export/. scripts/build_visuals.py generates SVG/PNG from JSON. Mermaid changes require matching JSON updates. These are architecture illustrations, not automatically generated UML models. Rebuilding requires Python, Pillow and Arial; the script specifies the macOS font path.

8. Key Workflow Mockups

DIP-10. 20 October 2026. Three schematic mockups, not a working ML service. Result values are illustrative; no model was run. No patient image or artificial Grad-CAM map has been inserted.

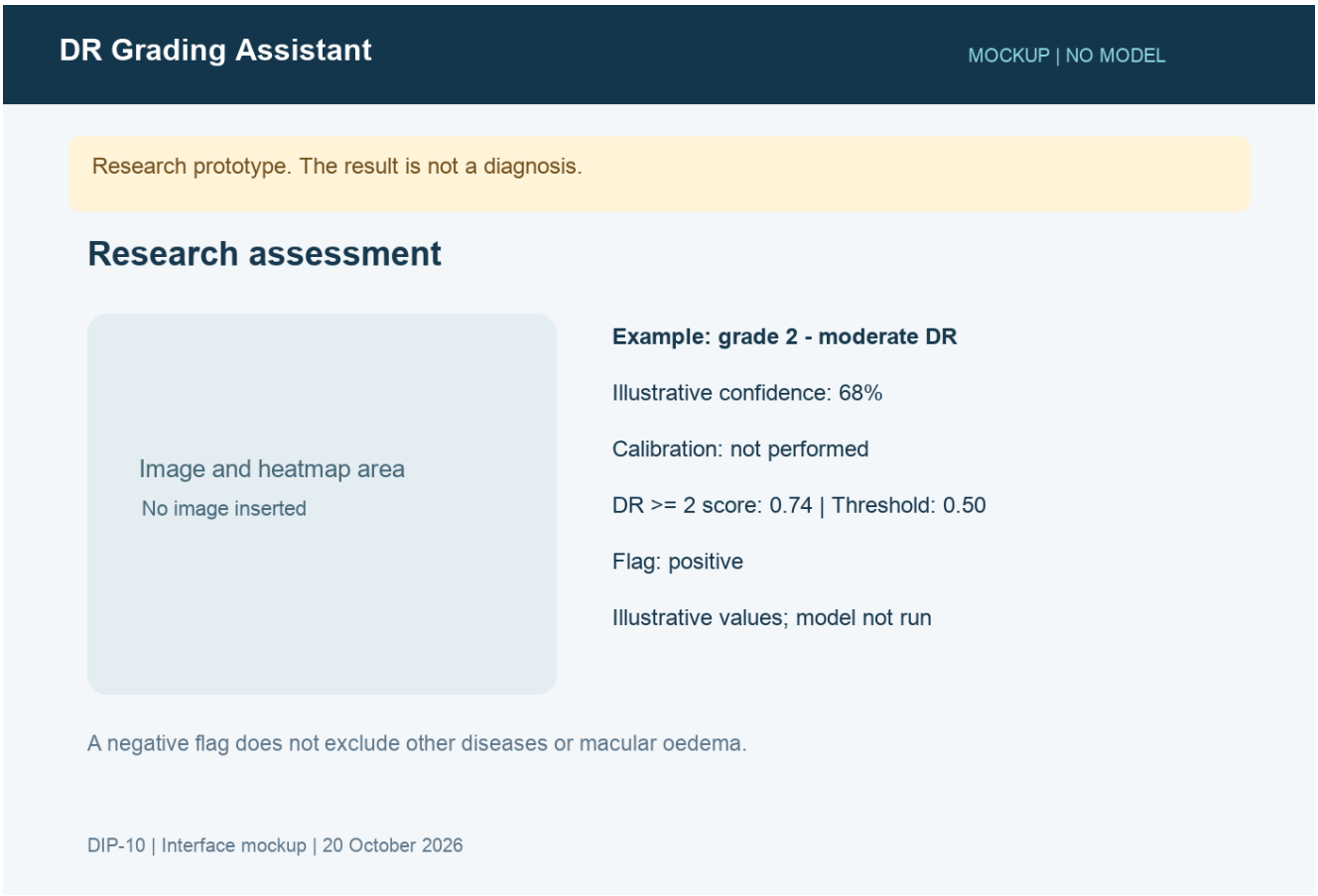
Upload



Upload

One file, 20 MiB limit and explicit research purpose. After selection, display the filename and replacement option; disable submission during analysis. Processing animation is a state of the future implementation.

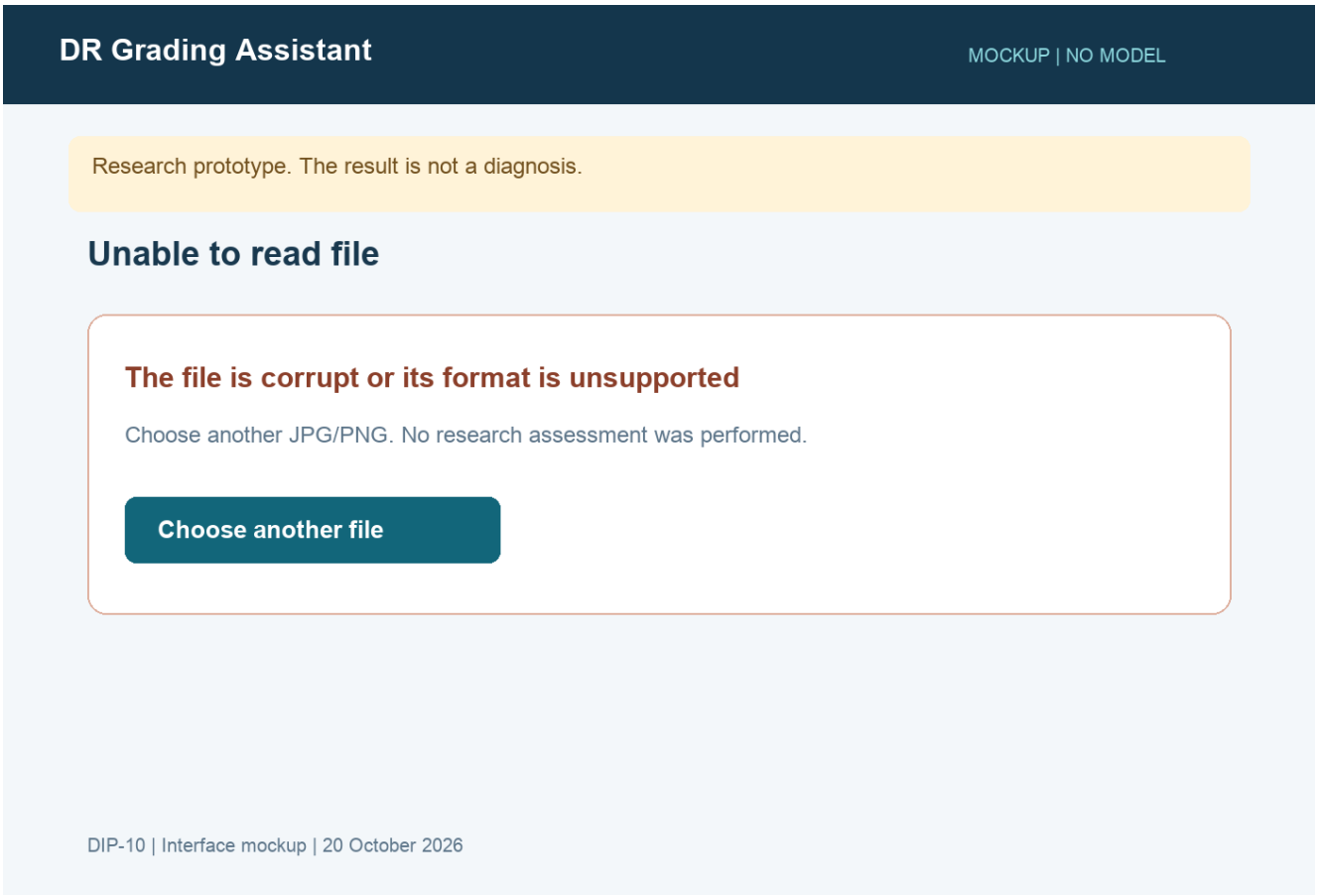
Result



Result

Show grade 0-4 separately from binary score. In the example, score=0.74 and illustrative threshold=0.50, so the flag is positive. The operational threshold will be selected on calibration only. Illustrative probabilities are [0.06, 0.20, 0.68, 0.04, 0.02], summing to 1. Implement map/original switching under FR-06. Confidence is model output, not the probability of being healthy. A negative flag does not exclude other diseases.

Error



Error

A corrupt file receives no prediction; select another image. Implementation must additionally handle size limits, unavailable models and internal failures. A decoding error is not a clinical image-quality assessment.

Mockup Verification

Scenarios map to UC-01/02/03 and FR-01..08. User validation has not occurred. Next, assess understanding of the binary flag, numerical confidence and limitations. Test keyboard access and 375 px responsiveness after implementation. Reproduce exports with scripts/build_visuals.py.

9. Verification and Evaluation Plan

Version 0.2, 20 October 2026. DIP-11. Experiment plan; no training, measured metrics or user trials have been completed in this version.

Data and Leakage Prevention

The first baseline uses APTOS 2019. EyePACS is an extension after access and rights checks. Reserve IDRiD and Messidor-2 for external evaluation; do not use them to select preprocessing, weights, thresholds or temperature. Provenance and conditions appear in docs/data-plan.md.

E01: build a manifest containing source, image_id, patient_id if available, label, SHA-256, dimensions and split. Detect identical files using SHA-256 and visually similar files using perceptual hashes and manual inspection. Group both eyes and visits of the same patient. Form groups first, then stratify into train 70%, tuning 10%, calibration 10%, test 10%; seed=42. If patient counts are small, document any ratio change before measuring metrics. If patient_id is unavailable, do not claim patient-level splitting; disclose potential performance inflation.

E02: freeze internal-test and external manifests before training. Derive preprocessing, class weights and samplers from train only. Use tuning for hyperparameters and early stopping, calibration for temperature and threshold, never weights. External sets must not influence selection. Check overlap with training sources before external evaluation. External testing is not clinical validation.

Baseline and Comparisons

E03: a naive baseline predicts training-class frequencies. The CNN baseline is ImageNet-pretrained EfficientNet-B0, 384x384, cross-entropy, AdamW, up to 30 epochs, early stopping on tuning QWK with patience=5. Initial lr=1e-4, weight_decay=1e-4, batch=16 or maximum that fits memory; record changes. Apply augmentation to train only; no random test transforms. Compare ResNet50 on the same split/resolution using seeds 42/43/44. Record actual time, memory, versions and cost.

Change one factor at a time: weighted loss, resolution or ordinal formulation. Compare against the fixed baseline; derive class weights from train only. Limit experiments to available resources; first complete one reproducible baseline. RETFound is a literature alternative, not mandatory implementation.

Metrics and Uncertainty

E04: the primary metric is quadratic weighted kappa for ordered classes 0..4; also report macro-F1 and the confusion matrix with counts. For binary evaluation, ground truth is y>=2 and score=p2+p3+p4. Measure AUROC and AUPRC. Select the threshold on calibration only: among thresholds with specificity >= 0.90, maximise sensitivity; break ties by higher specificity. Define threshold enumeration and equal-score handling in code. On test, report actual sensitivity/specificity at that same threshold; do not assume specificity remains 0.90. Argmax grade and the binary flag can disagree because they answer different questions.

Fit temperature T>0 on calibration by multiclass NLL. Report NLL, Brier score and top-label ECE before/after calibration using 15 equal-width bins: ECE = sum of bin fraction multiplied by |accuracy - mean confidence|. Show a separate reliability diagram and binary ECE for the binary score. ECE depends on binning and does not guarantee reliability of individual answers.

E05: estimate 95% percentile bootstrap confidence intervals using 2,000 patient resamples. If patient_id is unknown, resample images and disclose the limitation. For AUROC, skip and count degenerate resamples containing only one class. Compare models with paired bootstrap on the same observations; distinguish seed variation from sampling confidence intervals. Report separate IDRiD/Messidor-2 metrics, change relative to internal test, class distributions and cameras if known. Document label mapping before evaluation; do not invent five-grade labels. Omit metrics without suitable ground truth.

Prototype Tests

ID	Test	Expected result
T01	Valid JPG/PNG and grayscale	200 and valid contract
T02	Empty file, disguised extension, corrupt bytes	415 or 422 according to cause; no inference
T03	>20 MiB, >40 million pixels, filename containing ../	413 for limit; filename does not control path
T04	Fixed test logits	Finite softmax, sum 1; grade 0..4
T05	Scores below, above and equal to threshold	flag exactly matches score >= threshold
T06	Missing weights, exception, new upload	503/500 with request_id; no stale output; temporary files deleted
T07	Grad-CAM on control image	Correct geometry; map failure preserves main result
T08	Inspect every screen	Research-use and limitation statements
T09	Re-evaluate identical data/weights	Versions, manifest and comparable result retained
T10	Inspect logs/temporary directory	No images or personal information
P01	100 sequential requests after 10 warm-ups	Record p50/p95, errors, hardware; compare with 10 s target
U01	Keyboard and 375 px screen	All actions reachable, clear errors, no horizontal scrolling

Explainability Evaluation

Select correct and incorrect examples from every class, including overconfident errors. Generate Grad-CAM for the predicted class and compare with occlusion of salient regions. Discuss maps with the adviser after obtaining consent. Qualitative judgement does not replace quantitative evaluation or prove map causality [L04].

Stage Completion Criterion

Retain code, manifests, configurations, weights with checksums, reports of all executed experiments and limitations. Report negative results as well. QWK >= 0.85 is achieved only if measured on the actual internal test set; it is not a clinical-use threshold.

10. Project Management and Risks

DIP-12. 20 October 2026. Sole project owner: Roman Vladimirovich Kutbeyev (GitHub: lengway). Supervisor: Marzhan Kabdykalievna Turarova.

Work Tracking and Evidence

Repository: https://github.com/lengway/diabetic-retinopathy-grading . YouTrack: https://diabeticretinopathygrading.youtrack.cloud . Project DIP uses Kanban with the Stage field. Each issue specifies an assignee, deliverable, acceptance criteria and due date. Prepare changes in a DIP-ID branch, retain real commit dates and use Pull Requests. Do not rewrite history to imitate gradual work. Set Done only after checking criteria.

The sole student's contribution covers problem formulation, analysis, requirements, architecture and UX, followed by data/ML, service and testing. This version does not claim completed training or implementation. Match contribution statements to files/Git history before submission. Earlier verified work includes DIP-1 scaffolding, DIP-2 proposal and DIP-12 planning/YouTrack link.

Immediate Schedule

Date	Deliverable	Dependency
20 October	Reviews, comparisons, requirements, architecture, mockups and draft report	Primary sources
21 October	3-5 user interviews or school-accepted client evidence; feedback request	Specialist availability and format agreement
22 October	Authorised dataset acquisition and EDA; review corrections	Accounts/dataset terms
23 October	Documents, actual statuses, committee access and screenshots	Recipient account information
24 October, by 17:00	Final check and upload of two PDFs by one representative	University account
24 October, 18:00	School form closes	Astana time / Asia/Almaty

The one-hour reserve is internal; the official deadline is 18:00 on 24 October 2026. Completing all interviews within one day is not guaranteed and remains the main open risk.

Thesis Milestones: Proposed Schedule

November-December 2026: data and baseline. January-February 2027: external evaluation, calibration and explainability. March-April: service and tests. April-May: thesis. Confirm defence dates with the school. Create pre-defense-1, pre-defense-2 and final-pre-defense tags at actual checkpoints; do not move them.

Risk Register

ID	Risk	Likelihood / impact	Mitigation and trigger	Owner
R1	User need unconfirmed	High / high	Schedule interviews; report missing records by 22 October	Student
R2	No supervisor feedback	High / high	Prepare a specific review package; record approval only after response	Student
R3	Committee cannot access private materials	Medium / high	Identify accounts, grant read access and verify as recipient	Student
R4	Licence/label restrictions	Medium / high	Rights register before download; no image redistribution; document dataset changes	Student
R5	Eye/patient/duplicate leakage	High / high	Grouping, hashes, split tests; disclose unknown patient_id	Student
R6	External performance drop	High / high	Untuned external test, report drop and limit conclusions	Student
R7	Imbalance and weak classes	High / medium	Per-class recall, macro-F1; train-only weighted loss	Student
R8	Insufficient GPU/time	Medium / medium	Start with B0; limit experiments; agree budget before renting	Student
R9	Overconfidence or diagnostic reading of maps	Medium / high	Calibration, error analysis, clear research-use labels	Student
R10	Form unavailable near deadline	Medium / high	Sign in early, check file sizes/fields and retain submission confirmation	Student

Access

The repository is private. As of 20 October 2026, the owner has accepted access and one invitation is pending. Committee access has not yet been granted.

11. Supervisor Feedback

Accepted. Work is in progress.

Data Plan and Access Conditions

DIP-6. 20 October 2026. Datasets were not downloaded in this session; EDA has not been completed. Figures describe published sources/plans, not local-file measurements.

Dataset	Purpose	Primary source	Restriction
APTOS 2019	Initial train/tuning/calibration/test	https://www.kaggle.com/competitions/aptos2019-blindness-detection/data	Check/accept competition rules in own account; availability is not redistribution permission
EyePACS/Kaggle 2015	Additional training	https://www.kaggle.com/competitions/diabetic-retinopathy-detection/data	Separate rules, volume and patient identifiers; check before combining
IDRiD	External evaluation	https://idrid.grand-challenge.org/ ; L07	Different subtasks; disease-grading images/labels are required, not masks alone
Messidor-2	External evaluation	https://www.adcis.net/en/third-party/messidor2/	Images and third-party labels may have different terms; confirm label source and mapping

Before downloading, record date, version, usage terms, registration, training permission, image/weight redistribution and required citation. Rights status remains 'unconfirmed' until checked. Clinical patient data are outside current scope.

EDA must report actual file/missing/duplicate counts, class distribution 0..4, dimensions, decoding failures, class examples where publication is permitted, patient_id limitations and split protocol. If access is unavailable, leave the task open; do not substitute published frequencies for local analysis. Do not commit datasets or weights to Git.